

Ben Finkelshtein

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Education

University of Oxford

January 2023 - Present

Doctor of Philosophy (PhD) in Computer Science

Oxford, United Kingdom

- Titled "Structure-based Learning in Deep Graph Networks"; how to utilize structure in Geometric Deep Learning
- Supervised by DeepMind Chair of AI Prof. Michael M. Bronstein and Dr. İsmail İlkan Ceylan
- Co-authored Towards Data Science's [Graph & Geometric ML in 2024](#) and [Cooperative Graph Neural Networks](#)
- Awarded the **Clarendon Scholarship of over £200,000** towards tuition fees and living costs (< **1% acceptance**)

Technion - Israel Institute of Technology

November 2019 - June 2022

Master of Science in Computer Science

Haifa, Israel

- Supervised by Prof. Alexander Bronstein and Prof. Chaim Baskin
- **GPA: 99.1; Graduated with highest honors (summa cum lauda, top 1%)**

Technion - Israel Institute of Technology

November 2013 - July 2017

Bachelor of Science in Electrical Engineering and Bachelor of Science in Physics

Haifa, Israel

- Completed two separate B.Sc as a Fellow in the IDF's competitive Psagot program (< **1% acceptance**)
- **GPA: 92.1; Graduated with great honors (magna cum lauda, top 3%)**

Selected Publications

Author of [Covered Forest: Fine-grained generalization analysis of graph neural networks](#)

Under Review

Expanding graph similarity theory to evaluate how graph structure and aggregation impact model generalization

First author of [Learning on Large Graphs using Intersecting Communities](#)

NeurIPS 2024

A fundamentally different pipeline for learning on very large non-sparse graphs using intersecting communities

First author of [Cooperative Graph Neural Networks](#)

ICML 2024

A new graph signal processing paradigm in which each node can choose a different communication strategy

Co-first author of [Strategic Classification with Graphs Neural Networks](#)

ICLR 2023

Learning in a setting where users that are dependant can modify their features to obtain favorable predictions

First author of [A Simple and Universal Rotation Equivariant Point-cloud Network](#)

ICML 2022

A simple architecture which is equivariant to rigid motions with the ability to approximate any equivariant function

Professional Experience

eBay; Applied Researcher

June 2022 – January 2023

- **Engineered LLMs** to match queries with the most relevant landing pages for search engine optimization
- Developed a keyword extraction technique, combined it with **GPT3**, and **subsequently fine-tuned the process**
- Achieved a 30% increase in landing page click-through rates (CTR) within the first three months of implementation

SKF Group - AI Center of Excellence; Data Scientist

August 2020 – May 2021

- Designed, customized, and implemented end-to-end machine learning pipelines, from ideation to production
- Led research in event prediction, **anomaly detection**, and **time series forecasting** to aid machinery fault diagnosis
- Designed an event-based evaluation metric that improved the company's anomaly detection from 78% to 90%

Rafael Advanced Defense Systems; Algorithm Developer

October 2017 – July 2020

- Specialized military service assignment (**1 recruit per year**)
- **Created a neural network which predicts n -body problem solutions** for classified company use
- Engineered generative models to predict molecular dynamics pathways, utilizing physical simulations
- Created new **generative models to analyze stable & transient protein folding conformations**

Awards

G-Research Grant - £2000	September 2024
Graduated Top 1% in Master of Science in Computer Science - £500	July 2023
Clarendon Scholarship - over £200,000	July 2022
M.Sc Study Scholarship - over £45,840	December 2019
Graduated Top 3% in Bachelors of Science in Electrical - £100	July 2018
Graduated Top 3% in Bachelors of Science in Physics - £100	July 2018

Selected Talks

Institute of Science and Technology Austria ; Graph & Geometric ML in 2024	November 2024
Technion - Israel Institute of Technology, CS department ; Graph & Geometric ML in 2024	September 2024
Technion - Israel Institute of Technology, ECE department ; Cooperative Graph Neural Networks	September 2024
RWTH Aachen University; Cooperative Graph Neural Networks	July 2024
Learning On Graphs, Oxford ; Cooperative Graph Neural Networks	December 2023
University of Oxford ; Graph Neural Networks meet Reinforcement Learning	June 2023
Technion - Israel Institute of Technology; Simple & Universal Rotation Equivariant Point-cloud NNs	January 2022
SKF Group - AI Center of Excellence; An Introduction to Geometric Deep Learning	April 2021
KDD Conference - Deep Learning on Graphs Workshop; Single-Node Attack for Fooling GNNs	August 2021

Teaching Experience

University of Oxford; Head Teaching Assistant January 2023 – Present

- Served as head TA for the **Geometric Deep Learning and Graph Representation Learning** courses
- Created the coding assignments for the different classes
- Guided 30 graduate students through coursework, while in charge of 100 students
- Led 10 teaching assistants, coordinating effective tutorial sessions, and timely support for students

Blue Education; Private Tutor July 2023 – Present

Taught Math, Physics and Computer Science for A-Levels, SAT, and college exams as an online part time tutor

Technion - Israel Institute of Technology; Head Teaching Assistant March 2021 – June 2022

- Created all of the [tutorials](#) and [code notebooks](#)
- **Conducted hybrid online and classroom lectures** and practice sessions for the Geometric Deep Learning course
- Supervised 40 graduate and undergraduate students including final course projects

Matematix - High-School and College Exam Preparation Center; Teacher July 2012 – November 2013

- Led sessions of thirty students twice a week
- **Taught Mathematics, Physics, English and Coding for matriculation, SAT, and college exams**

Technologies, Languages and Interests

Technologies: C#, C++, Python (Pandas, Numpy, Pytorch, Cuda, Pytorch geometric, Scikit-Learn), Matlab, SQL, Bash, Scala, LaTeX, Ubuntu, Git/version control, AWS, VMs, GNU/linux

Languages: English (Fluent), Hebrew (Native) and Russian (Basic)

Interests: Geometric Deep Learning, Large Language Models, Generative Models and Machine Learning Theory; Bachata dancer and instructor, Billiards and Powerlifting